

Operations & Deployment Runbook — NextGen ERP

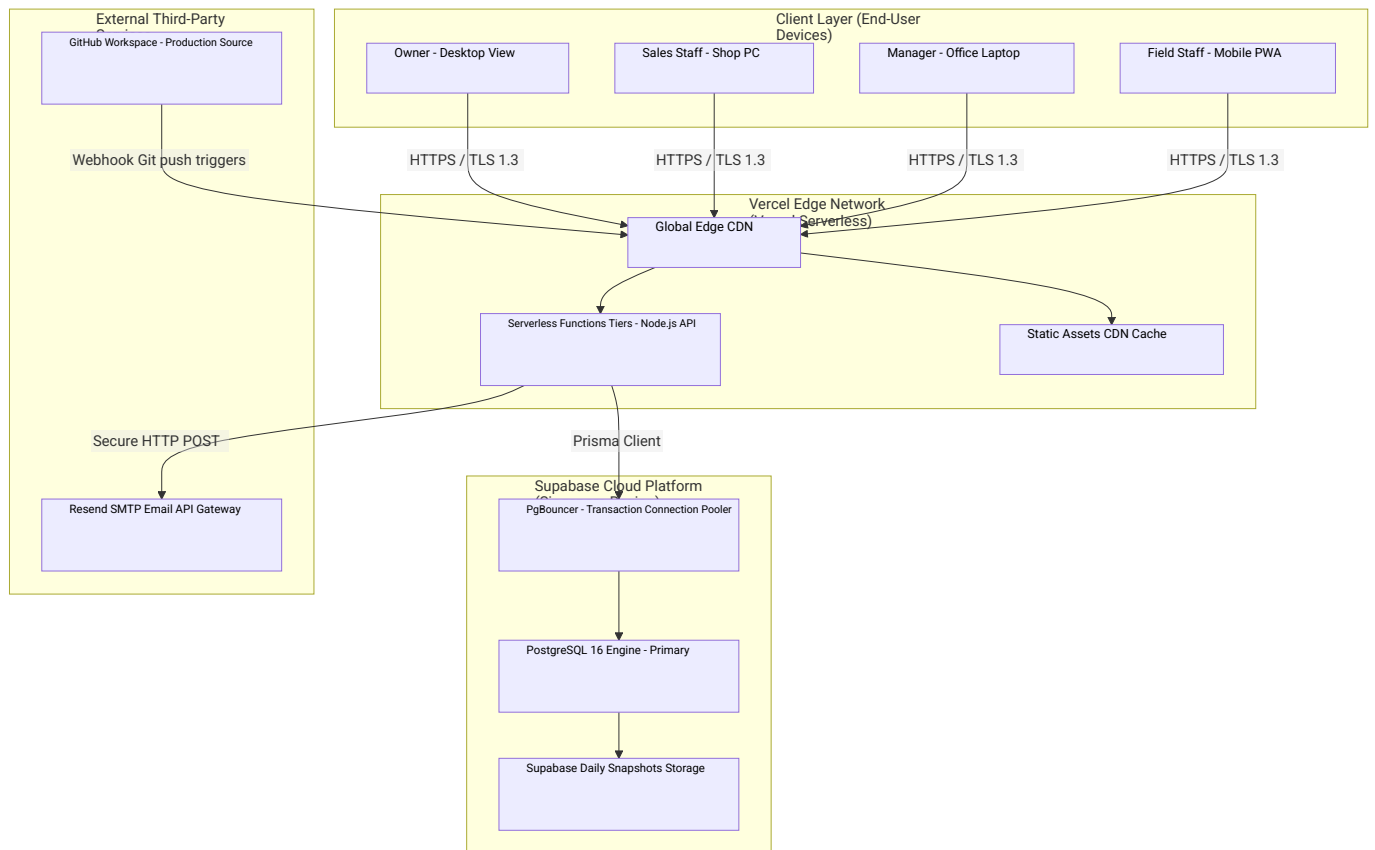
SECTION 1: System Overview

NextGen Interior And Waterproofing ERP is a cloud-native platform distributed across fully managed infrastructure tiers to ensure maximum availability, automated scalability, and zero maintenance overhead for the local business staff.

Production Service Coordinates:

- **Application URL:** <https://nextgen.com.np> (Primary production domain).
- **Presentation & Server Actions Tiers:** Vercel Serverless Hosting Engine.
- **Relational Database:** Supabase PostgreSQL 16 cluster, deployed in the Singapore (ap-southeast-1) region to achieve low latency from Nepal.
- **Transactional Mail Gateway:** Resend API, handling automated OTP account recoveries.
- **Source Code Repository:** Hosted on private GitHub workspaces.

Production Deployment Topology:



SECTION 2: Deployment Pipeline

The codebase enforces a multi-branch deployment flow to protect the live production system from development bugs. We implement three target branches: develop (local dev work), staging (pre-production testing), and main (live environment).



Detailed Deployment Instructions:

1. Developing a New Feature:

- Developers must create a separate branch off develop: `git checkout -b feature/quotations`.
- Run static checks before pushing: `npx tsc --noEmit && npm run build`.
- Push changes: `git push origin feature/quotations` and open a Pull Request (PR) to develop.

2. Pre-production Staging Deployment:

- Once merged into develop and verified, merge develop into staging.

- Vercel automatically detects the push and deploys the preview build.
- The QA team performs Client User Acceptance Testing (UAT) on this staging preview URL.

3. Production Deployment (Main Release):

- Merge the approved staging branch into main.
- The production runner compiles components, executes Prisma database structure migrations (npx prisma db push / prisma migrate deploy), and updates live routing on the production CDN.

4. Rolling Back a Bad Release:

- If a critical bug bypasses QA, log into the Vercel Dashboard -> go to **Deployments** -> locate the previous successful deployment card -> click the three dots -> select **Promote to Production**. This rolls back the live server to the previous stable state within 5 seconds without changing the git history.

SECTION 3: Environment Variables Reference

These variables are defined securely inside Vercel's encrypted Settings panel and locally inside .env.local (never committed to git).

Variable Name	Environment	Description & Source	Rotation Frequency
DATABASE_URL	Vercel + Local	Supabase -> Settings -> Database. The direct transaction pooler connection string.	Only if credentials are leaked.
NEXTAUTH_SECRET	Vercel + Local	Cryptographic string generated using openssl rand -base64 32 to sign cookies.	Annually.
NEXTAUTH_URL	Vercel + Local	The canonical live domain URL (e.g. https://nextgen.com.np).	If company domain is changed.
RESEND_API_KEY	Vercel + Local	Resend Dashboard -> API Keys. Used to authorize email recovery dispatches.	Only if leaked.

NEXT_PUBLIC_VAT_RATE	Vercel + Local	Holds default standard VAT percentage (13).	If Nepal national tax rate changes.
----------------------	-------------------	--	--

Impact of Missing/Wrong Variables:

- **DATABASE_URL:** App completely crashes; users see a 500 Server Error on all dynamic routes.
 - **NEXTAUTH_SECRET:** Users cannot log in; all password forms fail to generate session cookies.
 - **RESEND_API_KEY:** Staff account recovery OTP mails fail to send; logs report mail delivery errors.
-

SECTION 4: Database Backup & Recovery

Because financial accounting requires zero data loss, the backup and disaster recovery runbook must be strictly executed by the systems administrator.

4.1 Automatic Cloud Backups (Supabase):

- **Free Tier Configuration:** Supabase executes a full physical snapshot of the database once a day, retaining the backups for **7 days**.
- **Pro Tier Option:** Retains hourly logical snapshots for **30 days** and supports Point-in-Time Recovery (PITR) to restore the database to a specific second.
- **How to Restore:** Go to Supabase Dashboard -> **Database** -> **Backups** -> select the target date and time -> click **Restore Backup**.

4.2 Manual Database Backups:

To create a hard copy on your local storage before major upgrades, use our custom shell utilities:

```
# 1. Run the built-in backup script (configured in package.json)
npm run backup
```

```
# 2. The script executes a pg_dump stream and saves a compressed file to:
# /nextgen-erp/backups/nextgen_erp_YYYY-MM-DD.sql.gz
```

```
# 3. To manually restore the database from a backup file:  
gunzip -c nextgen_erp_YYYY-MM-DD.sql.gz | psql $DATABASE_URL
```

4.3 Mandatory Backup Triggers:

You must trigger a manual backup immediately before:

- Running Prisma schema changes (npx prisma db push or schema edits).
- Running database data repair migrations or batch SQL routines.
- Major dependency package updates.

4.4 Emergency Disaster Recovery Procedure (Database Corruption):

1. **Detect:** System logs report database errors, or users report total loss of record visibility.
2. **Alert:** Promptly notify Nischal Timsina (Business Owner) of the system downtime.
3. **Isolate:** Place Vercel into **Maintenance Mode** by rerouting traffic to a static page or pausing Vercel serverless functions, blocking incoming data requests.
4. **Inspect:** Open the Supabase Database dashboard to verify project status. If the DB is healthy but data is corrupted, select the most recent automated daily backup.
5. **Restore:** Trigger the database restore procedure from the Supabase Backups dashboard.
6. **Verify:** Run query checks on SalesInvoice and LedgerEntry tables to ensure cash balances and invoices are complete up to the restore point.
7. **Resume:** Remove the Vercel maintenance block, resume full client routing, and publish an incident wrap-up note.

SECTION 5: Monitoring & Alerting

To maintain a healthy platform, administrators must review service metrics on a regular operational schedule.

5.1 Service Metrics Thresholds:

Metric	Where to Check	Warning Threshold	Operational Remedy
Application Errors	Vercel -> Functions ->	Any recurring 500 status codes.	Inspect error traceback for code typos or bad database connection

	Logs		pools.
Database Storage	Supabase -> Settings	400MB (80% of Supabase 500MB free limit).	Upgrade to Supabase Pro Tier or run database storage cleanup routines.
Email Sends	Resend.com -> Logs	2,800 emails/month (near 3,000 free limit).	Upgrade Resend tier to avoid locking staff account recoveries.
Build compilation	GitHub -> Actions tab	Any red (FAILED) build card.	Review commit log details for TypeScript declaration or build syntax mismatches.

5.2 Common Production Incidents & Quick Fixes:

- Incident: Login screen returns a "500 Server Error" on password verification.
 - **Check:** Verify the NEXTAUTH_SECRET environment variable in the Vercel settings panel.
 - **Fix:** Re-generate a secure 32-character key, update the Vercel variable value, and trigger a redeployment.
- Incident: Database Connection Refused or Paused.
 - **Check:** Free Supabase database instances pause automatically after 7 days of developer inactivity.
 - **Fix:** Go to the Supabase dashboard page and click the green **Unpause Project** button to resume services.
- Incident: Invoice download prints as a blank page.
 - **Check:** Vercel Serverless Logs for font compilation errors in @react-pdf/renderer.
 - **Fix:** Verify that custom printing fonts are correctly loaded as relative assets during the compilation build stage.

SECTION 6: Upgrading & Maintenance

6.1 Running Regular Database Schema Updates:

1. Open and edit /prisma/schema.prisma locally.
2. Apply changes to your local development environment: `npx prisma db push`.
3. Verify that the local Next.js compiler runs correctly: `npm run build`.
4. Commit changes: `git add prisma/schema.prisma && git commit -m "feat: add schema fields"`.
5. Upon merging staging -> main, Vercel's build runner will automatically execute the schema

changes using Prisma, preventing runtime table discrepancies.

6. *Caution: Never drop or rename columns containing historical data (e.g. customer balances) without first running a data migration script to preserve balances.*

6.2 Annual System Maintenance Checklist:

- **Rotate Credentials:** Renew the NEXTAUTH_SECRET and Resend keys to protect data security.
 - **Archive Logs:** Move audit logs older than 12 months into historical CSV exports, keeping the database snappy.
 - **Initialize Fiscal Year:** Go to **Settings** -> **Fiscal Years** and create the new Shrawan-Ashadh calendar period. Set the new year as CURRENT and mark the completed year as CLOSED.
 - **Review Access Levels:** Clean up deactivated employees and verify active staff roles.
 - **Test Backups:** Trigger a manual restore to a local test database to confirm backups are fully functional.
-

SECTION 7: For the Business Owner (Non-Technical)

As the owner of NextGen Interior And Waterproofing, you retain full ownership and control of your system and data.

7.1 Support Contact Information:

- **Primary Developer:** Rabin Sharma (Jhapa, Nepal)
- **Email:** rabinsharma.dev@gmail.com
- **For Database Support:** Open a ticket at support.supabase.com.
- **For Email Delivery Support:** Visit the support section at resend.com.

7.2 What You Can Manage Without Developer Help:

- **Business Profile details:** Change your firm's name, PAN tax number, phone number, and physical office addresses directly in **Settings** -> **Business Profile**.
- **User management:** Add new employees, assign them roles (e.g. Sales Staff or Viewer), and deactivate accounts immediately if staff leave the company.
- **Data backups:** Access **Settings** -> click **Export Data** to download a compressed database backup anytime.

- **Audit reports:** Check all financial sheets and customer outstanding dues lists.

7.3 Understanding Monthly Operating Costs:

The ERP system runs on optimized, free tiers to keep your overhead costs at zero during normal operations.

Service	Monthly Cost	Included Free Limit	Upgrade Indicator
Vercel	NPR 0.00 (Free)	100GB monthly bandwidth	High site traffic spikes.
Supabase	NPR 0.00 (Free)	500MB database storage	Database approaches 400MB size.
Resend	NPR 0.00 (Free)	3,000 emails/month	Large team doing hundreds of recovery OTPs.
Domain (.com.np)	NPR 0.00 (Free)	Lifetime license	Never (permanent domain registration).

7.4 Data Ownership Guarantee:

Your business database is hosted securely inside the Supabase platform under your official corporate account credentials:

- **Primary Owner Account:** gauravchaulagain99@gmail.com
- **Data Export Policy:** You can download all transaction tables, client lists, and bookkeeping logs at any time. Your data belongs strictly to you.